Production Vacation-Rental Marketplace Architecture

Scaling Blueprint: Frontend & SSR • API Gateway & BFF • Microservices • Storage & CDC • Geo Search • AWS EKS Deployment

SCALE: 100M+ REQS/DAY

1. FRONTEND & EDGE CDN TIER

Client apps, Edge SSR, Stale-While-Revalidate & Asset Delivery

Your React App (App.jsx + PhotoTour.jsx)
Listing PDP, Dynamic Calendar, Guest Picker, Interactive Modal Tour
• Code-split PhotoTour.jsx via React.lazy() | Bundle: <120KB initial

Native Mobile Clients (iOS Swift / Android Kotlin)
GraphQL Mobile API Consumer with offline draft caching & push sync

Cloudflare Anycast CDN & Edge WAF
DDoS Shield • Edge HTML Cache (SWR 24h) • SSL Termination
Cloudflare Images / Lambda@Edge: On-the-fly WebP/AVIF generation
• 98.2% Cache Hit Ratio on static assets & pre-rendered PDP HTML

2. GATEWAY, BFF & SERVICE MESH

Traffic routing, GraphQL federation, auth, rate limiting & mTLS

AWS Network Load Balancer (NLB)
Ultra-low latency L4 Passthrough • Multi-AZ Anycast IP Routing

BFF & GraphQL Federation Gateway (Envoy / Viaduct)
• Schema Stitching: Combines Listing, Pricing, Host & Reviews into 1 query
• Auth: JWT verification, OAuth2 & Role-Based Access Control (Guest/Host)
• Distributed Rate Limiting: Redis sliding-window token bucket (per IP/user)

Istio / Envoy Service Mesh
mTLS Zero-Trust Encryption • Circuit Breaking • Outlier Detection • W3C Tracing

6. CLOUD DEPLOYMENT & OBSERVABILITY

Elastic Kubernetes, GitOps CI/CD, Canary & Telemetry

AWS EKS Multi-Region Kubernetes
Pod Autoscaler (HPA) via CPU & Custom RPS Metrics
• Karpenter: Provisions right-sized Graviton EC2 nodes in <45s

GitOps & Progressive Delivery (ArgoCD)
Canary rollouts (5% -> 25% -> 100%) with automated rollbacks
• Rollback threshold: >0.05% error rate or >15% P99 latency drift

Full-Stack Observability & SRE
OpenTelemetry Traces • Prometheus Metrics • Datadog APM • PagerDuty
• P99 Latency SLA: <150ms for PDP | Availability: 99.99%

3. CORE DOMAIN MICROSERVICES TIER (DOCKER CONTAINERS ON KUBERNETES EKS)

Domain-Driven Design (DDD) with decoupled services communicating via gRPC (synchronous) and Kafka (asynchronous)

Search & Ranking
Geo-proximity query (H3)
Two-stage ML candidate rank
Personalized guest scoring
< 35ms Geo Response

Listing Service
Feeds your App.jsx PDP
Amenity specs, house rules
Photo metadata & albums
Cached in Redis L2

Booking Engine
Saga Orchestrator (2-phase)
Distributed lock (Redlock)
Zero double-booking lock
Strict ACID Guarantees

Availability Service
365-day Date Bitmaps
O(1) Date range bitwise check
Instant date block/release
Sub-millisecond Checks

Pricing Engine
Seasonal surge, length discounts
Taxes, cleaning fee calculation
Multi-currency conversion
Rule + ML Price Model

Payments & Escrow
Stripe / Adyen integration
Double-entry ledger
Payouts on 24h post-checkin
PCI-DSS Compliant

4. EVENT STREAMING & CHANGE DATA CAPTURE (CDC)

Apache Kafka Distributed Event Spine
Topics: listing.updated • booking.reserved • payment.settled • review.created

Debezium Change Data Capture (CDC)
Non-blocking replication from Aurora Postgres WAL logs into Kafka (<200ms)

Apache Flink Real-time Stream Analytics
Real-time Search Index Sync • Fraud Velocity Detection • Dynamic Pricing Features

5. POLYGLOT STORAGE & SEARCH ENGINE CLUSTERS

Data storage architecture matched strictly to access patterns (Read/Write ratios, ACID vs Eventual Consistency)

Aurora PostgreSQL (OLTP)
Primary relational transactional store
Sharded by hash(listing_id)
Bookings, Hosts, Financial Ledgers
Multi-AZ Read Replicas (Auto-scale)
ACID • Row Versioning

Redis Cluster (L2 Distributed)
Calendar Availability Bitmaps (GETBIT)
Distributed Locking (Redlock Mutex)
Listing metadata cache (TTL 1 hr)
User session & active cart states
Sub-millisecond Latency

OpenSearch / Elasticsearch
Geo-distance & Bounding-box search
Sharded by Uber H3 Hexagonal Grid
Inverted index: Amenities, Price, Type
Vector index for semantic search
Near Real-Time (<500ms sync)

DynamoDB / Cassandra
High-write throughput key-value
Guest <-> Host Chat Messenger
Audit logs & user notification inbox
Idempotency request tokens
Unlimited Horizontal Scale

S3 Media + ClickHouse Lake
AWS S3 / Cloudflare R2 (Photos)
Pre-signed upload URLs for hosts
ClickHouse / Snowflake OLAP
Dynamic pricing training & reports
Petabyte Scale Analytics